

Isolating True Pitcher Talent: A Skill-based Framework for Measuring Eliteness

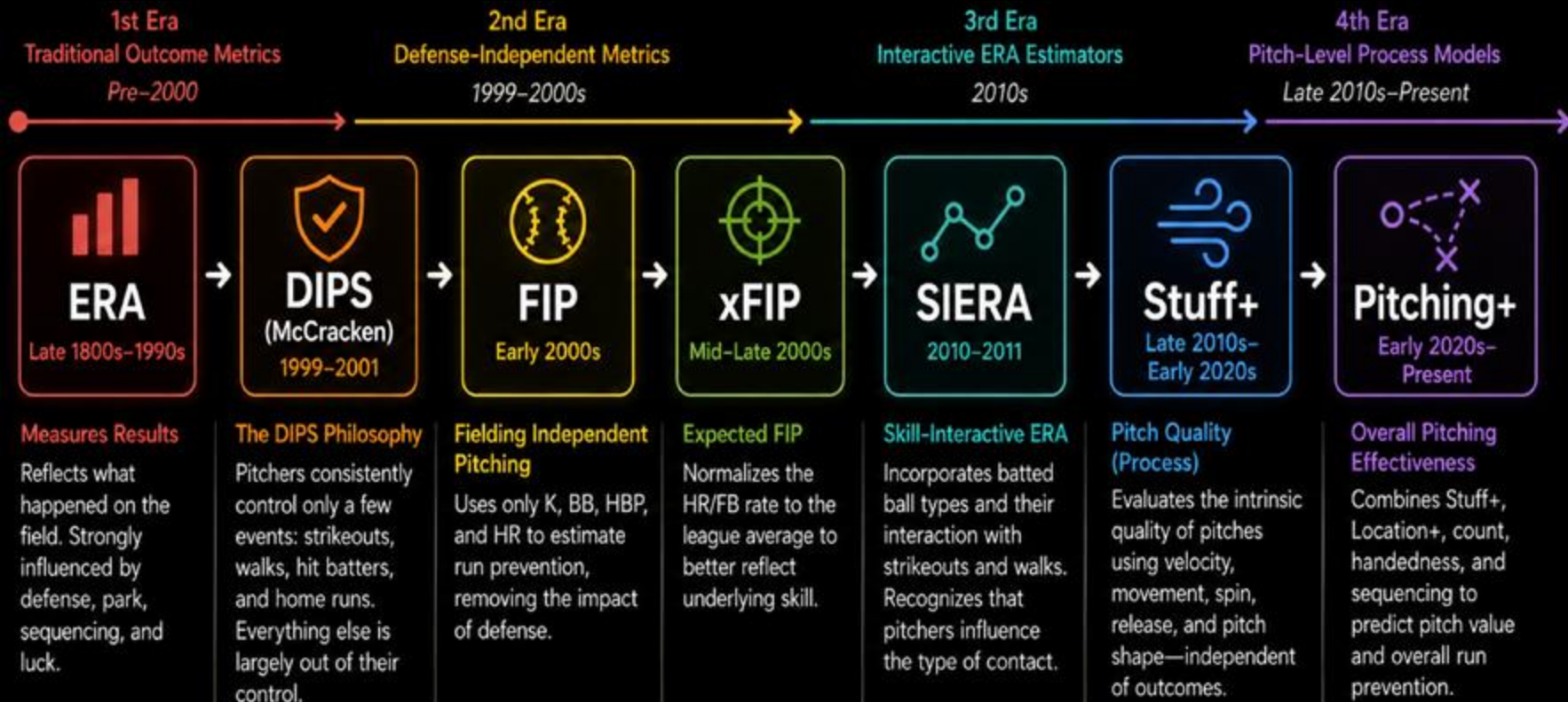


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Evolution of Pitching Evaluation

The future of pitching evaluation must measure true pitcher skills.



Results versus Skills

A pitcher's skill profile is latent given ERA and FIP.

PITCHER A



ERA 3.20

FIP 3.35

SKILL PROFILE

VELOCITY (STUFF) ELITE

SWING-AND-MISS (MISS ABILITY) ELITE

COMMAND AVERAGE

CONTACT SUPPRESSION AVERAGE

Similar Results



Different Skills

PITCHER B



ERA 3.18

FIP 3.40

SKILL PROFILE

VELOCITY (STUFF) AVERAGE

SWING-AND-MISS (MISS ABILITY) AVERAGE

COMMAND ELITE

CONTACT SUPPRESSION ELITE



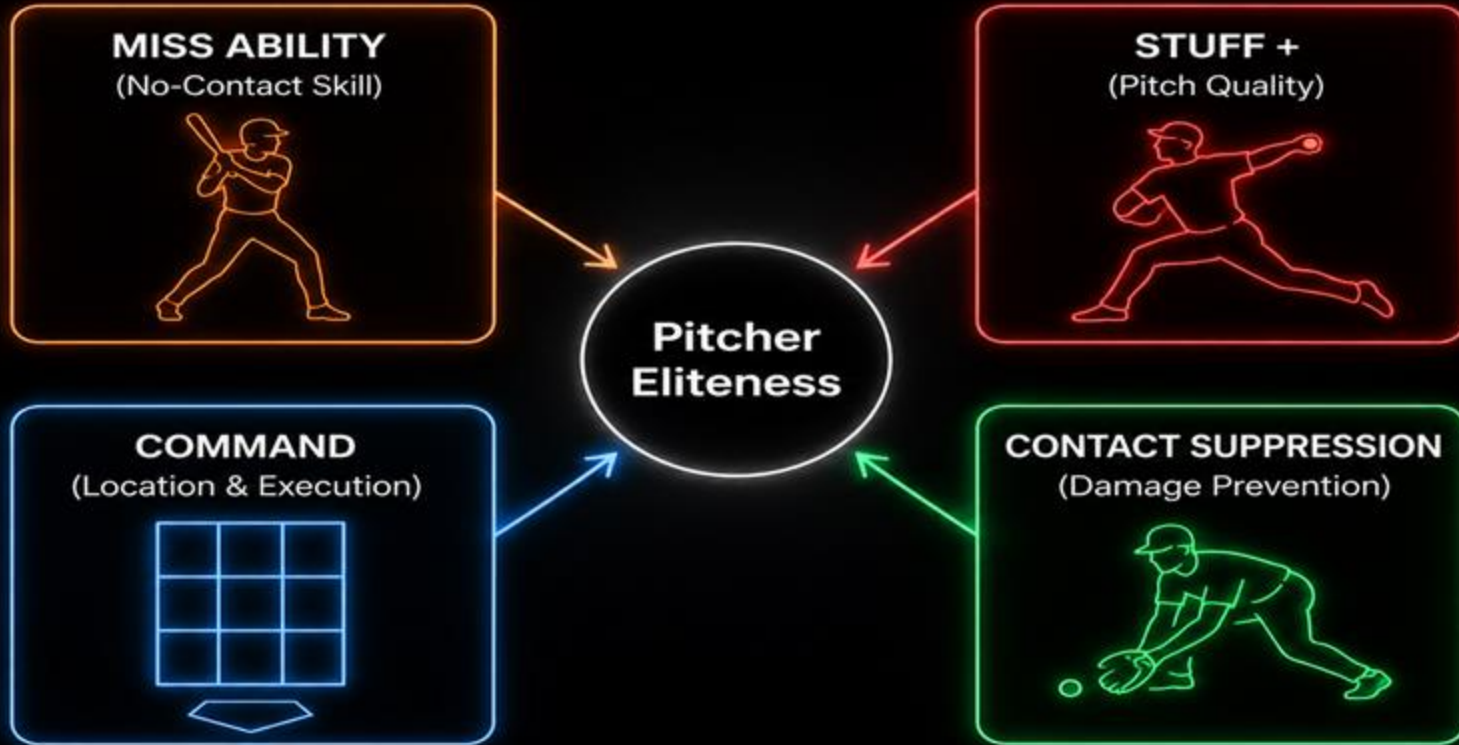
Traditional evaluation (ERA, FIP) tells us **what** happened.

But it does not explain **why** it happened—or whether it is **repeatable**.



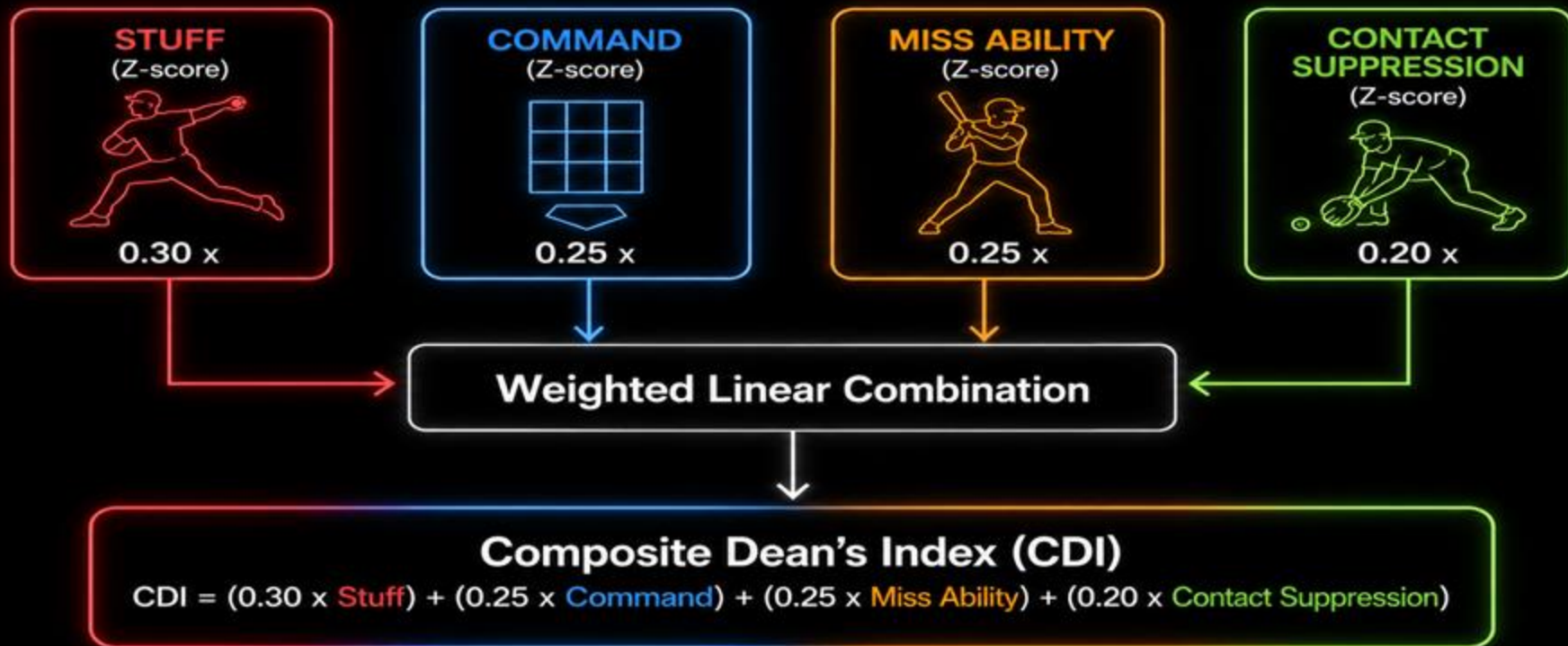
Pitching Eliteness Framework

Pitcher controlled variables must be used to measure true pitcher eliteness.



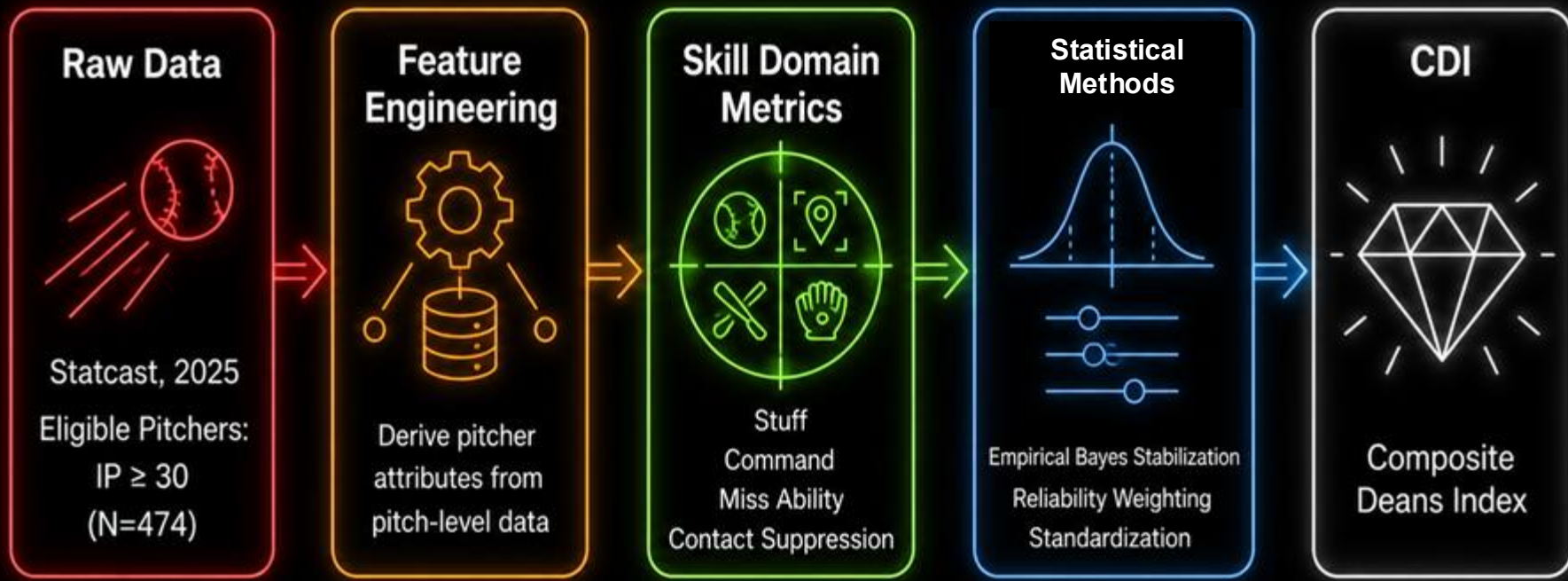
Composite Dean's Index

CDI is the future of measuring pitcher eliteness.



Research Approach

Research and development requires a substantive methodology.



Statistical Methodology

A model must be safeguarded from producing eronous results.



Empirical Bayes Stabilization

- Shrink noisy estimates toward league average
- Use opportunity counts by metric or pitch type
- Protects small samples from overreaction

$$\hat{\theta}_{EB} = \frac{n}{n+k} \hat{\theta} + \frac{k}{n+k} \mu$$



Reliability Weighting

- Higher sample size → higher trust
- Prevents unstable metrics from dominating
- Lets each skill carry evidence-based weight

$$S_i = \sum_{j=1}^P w_j z_{ij}, \quad w_j = \frac{r_j}{\sum r_j}$$



Standardization & Direction

- Convert stabilized metrics to z-scores
- Invert lower-is-better measures
- Combine comparable skill signals

$$z_{ij}^* = - \left(\frac{x_{ij} - \mu_j}{\sigma_j} \right)$$

MISS ABILITY: No Contact Skill

Quantifies a pitcher's ability to generate swings-and-misses and strikeouts independent of contact outcomes.



1. METRIC INPUTS

Miss Ability combines:

- CSW%
- Whiff%
- K%
- Put-Away Rate



2. STABILIZATION / SHRINKAGE

Each raw rate is stabilized toward the league average:

$$\text{Stabilized Metric} = \frac{n}{n+k} (\text{Pitcher Rate}) + \frac{k}{n+k} (\text{League Avg})$$



Rates with smaller samples are pulled closer to league average.



3. FINAL MISS SCORE

The final Miss Ability score is a reliability-weighted combination of four stabilized metrics:

$$\text{Miss Score} = \frac{W_{\text{CSW}} Z_{\text{CSW}} + W_{\text{Whiff}} Z_{\text{Whiff}} + W_K Z_K + W_{\text{PA2}} Z_{\text{PutAway}}}{W_{\text{CSW}} + W_{\text{Whiff}} + W_K + W_{\text{PA2}}}$$



Higher Miss Score =
stronger bat-missing profile



The Miss Ability score rewards pitchers who combine called strikes, swing-and-miss ability, strikeouts, and two-strike finishing skill while adjusting for sample-size reliability.

CONTACT SUPPRESSION: Damage Prevention

Measures a pitcher's ability to limit the quality of contact when the ball is put in play.



1. METRIC INPUTS

Contact Suppression combines:

- Avg Exit Velocity
- Hard-Hit%
- Barrel%
- Ground Ball%
- xwOBA
- xSLG



2. STABILIZATION / SHRINKAGE

Each raw rate is stabilized toward the league average:

$$\text{Stabilized Metric} = \frac{n}{n+k} (\text{Pitcher Rate}) + \frac{k}{n+k} (\text{League Avg})$$



Rates with smaller samples are pulled closer to league average.



3. FINAL CONTACT SCORE

The final Contact Suppression score is a reliability-weighted combination of six stabilized metrics:

$$\text{Contact Score} = w_{BIP} \cdot \text{mean}(z_{EV}, z_{HH}, z_{Barrel}, z_{xwOBA}, z_{xSLG}, z_{GB})$$



Higher Contact Score =
better damage prevention



The Contact Suppression score rewards pitchers who limit hard contact and extra-base damage, providing a clearer view of true run prevention skill.

COMMAND: Location & Execution

Measures strike-zone control, pitch placement accuracy, and the ability to expand the zone.



1. METRIC INPUTS

Command combines:

- Mean Plate Location (Horizontal & Vertical)
- Location Consistency (Horizontal & Vertical SD)
- Zone %
- Edge %
- Waste %
- Chase %



2. STABILIZATION / SHRINKAGE

Each raw rate is stabilized toward the league average:

$$\text{Stabilized Metric} = \frac{n}{n+k} (\text{Pitcher Rate}) + \frac{k}{n+k} (\text{League Avg})$$



Rates with smaller samples are pulled closer to league average.



3. FINAL COMMAND SCORE

The Command score is the average of eight standardized components:

$$\text{Command Score} = \frac{Z_{\text{MeanHor}} + Z_{\text{MeanVert}} + Z_{\text{SDHor}} + Z_{\text{SDVert}} + Z_{\text{Zone}} + Z_{\text{Edge}} + Z_{\text{Waste}} + Z_{\text{Chase}}}{8}$$



Higher Command Score = better location & execution profile



The Command score rewards pitchers who consistently hit their locations, work in the zone, limit waste, and induce chase swings—adjusted for sample-size reliability.

STUFF+: Physical Pitch Quality

External metric evaluating the physical characteristics of each pitch.

MODEL INPUTS

- Velo Difference
- Spin Rate
- VB Difference
- HB Difference
- Spin Efficiency
- HR Point
- VB
- Velo
- Spin Angle
- HB
- Extension

PITCH-TYPE MODELING

Physical Inputs



Separate model for each pitch type

FF

SI

SL

CH

CU



Pitch-type-specific weights



Pitch-type Stuff+ score

Each pitch type is evaluated independently,
and the inputs carry different weights by pitch type.

PITCHER-LEVEL STUFF+

Pitch-type Stuff+ scores



Weighted by pitch usage



Full pitcher Stuff+

$\text{Pitcher Stuff+} = \sum (\text{Pitch Usage} \times \text{Pitch-Type Stuff+})$

100 = League Average

Higher = Better



Rewards pitchers whose arsenals combine velocity, movement, spin, release traits, and pitch separation to create higher-quality offerings.

Outcomes

Different pitching skills drive different outcomes.



REGRESSION MODELS

MULTIPLE LINEAR REGRESSION (Full Skills Model)

$$Y = \beta_0 + \beta_1 \text{ Stuff} + \beta_2 \text{ Command} + \beta_3 \text{ Miss Ability} + \beta_4 \text{ Contact Suppression} + \varepsilon$$

Assesses the unique contribution of each skill while controlling for the others.

CDI (COMPOSITE) MODEL

$$Y = \beta_0 + \beta_1 \text{ Stuff (.30)} + \beta_2 \text{ Command (.25)} + \beta_3 \text{ Miss Ability (.25)} + \beta_4 \text{ Contact Suppression (.20)} + \varepsilon$$



$$Y = \beta_0 + \beta_1 \text{ CDI} + \varepsilon$$

Tests whether the weighted CDI alone predicts outcomes.

	 PitchBot xRV100 (Pitch Quality)	 SIERA (Run Prevention Skill)	2025  xERA (Expected ERA)	DEPENDENT VARIABLES (Outcomes)  xFIP (Expected FIP)	 FIP (Defense-Independent)	 ERA (Realized Outcome)	 RA9-Wins (Overall Value)
 Stuff	***	***	.	***	.	NS	.
 Command	***	***	***	***	***	**	***
 Miss Ability	**	***	***	***	***	***	***
 Contact Suppression	NS	***	***	***	***	***	***
 CDI (Composite Index)	***	***	***	***	***	***	***

*** $p < .001$; ** $p < .01$; * $p < .05$; . $p < 0.1$; NS Not Significant ($p \geq .05$)

Predictive Model

The CDI frames the future of predictive pitching evaluation.

 Outcome	 Correlation	 Cross-Validation R Squared	 Prediction Interpretation
 xERA	-0.753	0.715	 Excellent prediction of expected run prevention
 SIERA	-0.765	0.693	 Excellent prediction of future run-prevention skill
 xFIP	-0.677	0.567	 Strong prediction of defense-independent pitching ability
 PitchBot xRV100	-0.681	0.533	 Strong prediction of future pitch quality
 FIP	-0.653	0.508	 Strong prediction of defense-independent performance
 ERA	-0.534	0.383	 Moderate prediction of realized run prevention
 RA9-Wins	0.335	0.162	 Limited prediction of contextual value (team-dependent)

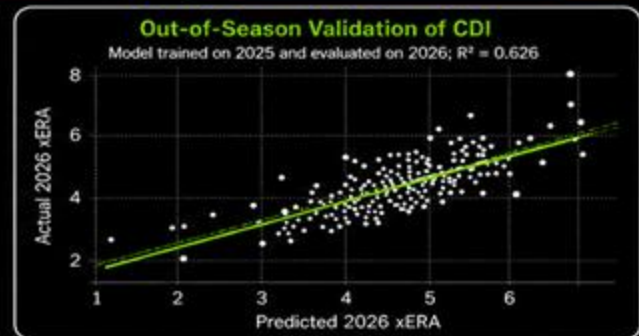
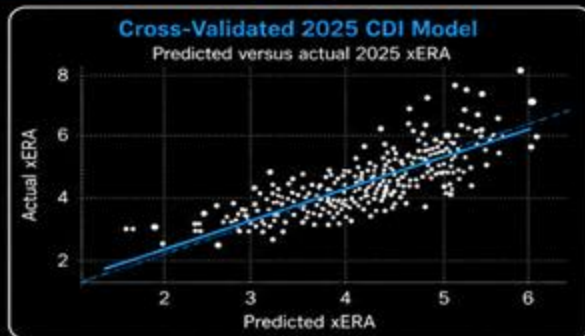
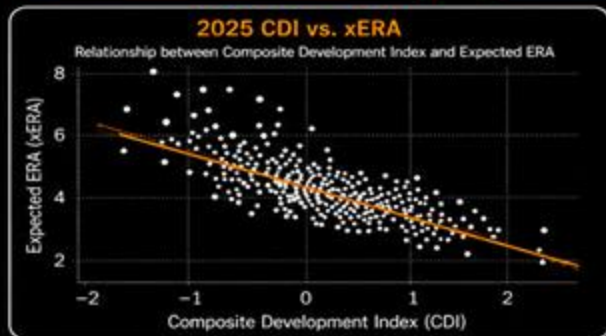
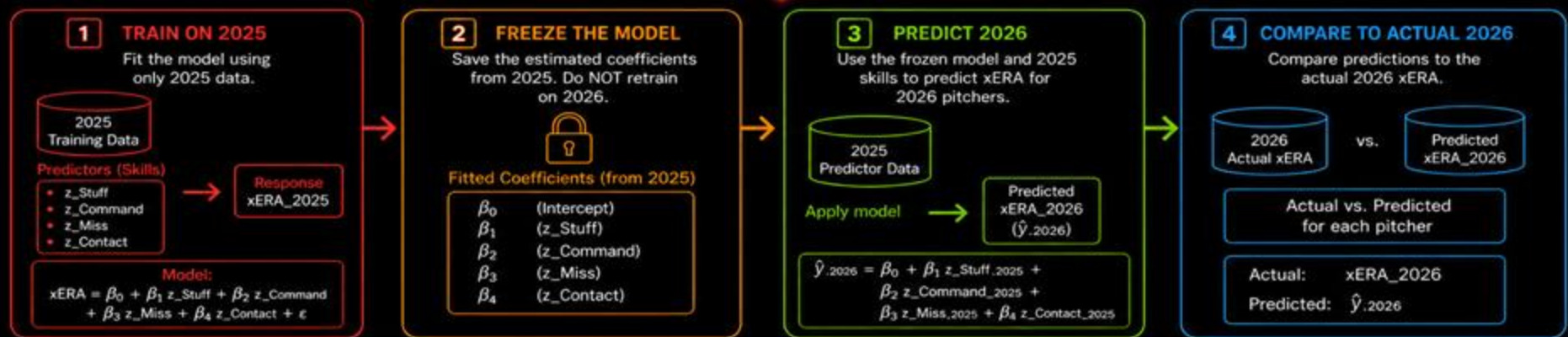


CDI excels at predicting skill-based outcomes (**xERA**, **SIERA**, **xFIP**, **PitchBot xRV100**, **FIP**), while traditional **ERA** is moderate and overall value (**RA9-Wins**) is context-dependent.



Machine Learning Prediction

Current pitcher skills predict tomorrow's xERA.



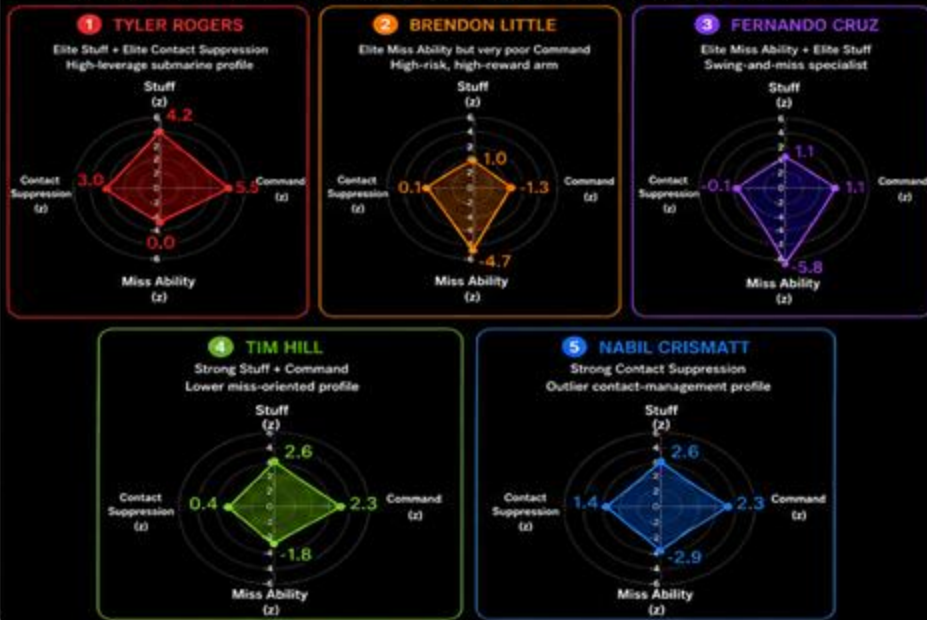
*** $p < .001$; ** $p < .01$; * $p < .05$; . $p < 0.1$; NS Not Significant ($p \geq .05$)

High-leverage Players in the Model

The model's results are not driven by extreme observation.

CDI Skill Fingerprints of the 5 Highest-Leverage Pitchers

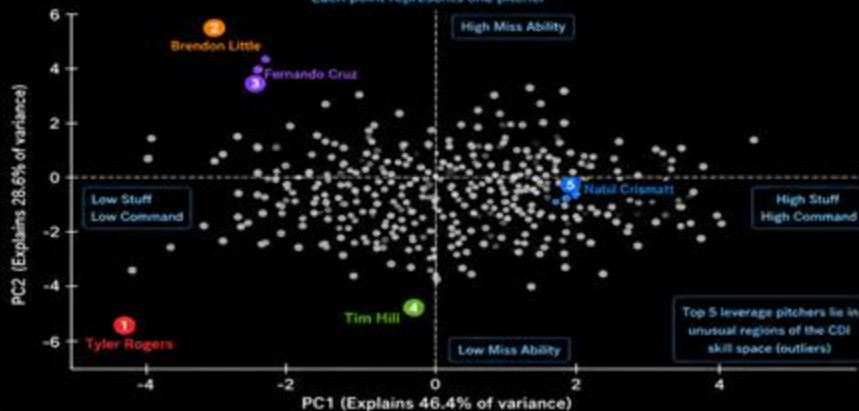
Standardized skill profiles from the 2025 ERA regression | Positive values indicate stronger CDI skill performance



These five pitchers occupy unusual regions of the four-dimensional CDI predictor space, but their low Cook's distances indicate that none exerted undue influence on the fitted model.

PCA Projection of the Four CDI Skill Domains (z-scores)

Each point represents one pitcher



Player	Stat Leverage (Leverage)	Cook's Distance (Influence)	CDI (Percentile)
1 Tyler Rogers	0.0845	0	100
2 Brendon Little	0.0711	0.0043	34.25
3 Fernando Cruz	0.0569	0.0048	61.95
4 Tim Hill	0.0531	0.0129	93.23
5 Nabil Crismatt	0.0434	0.0026	25.16

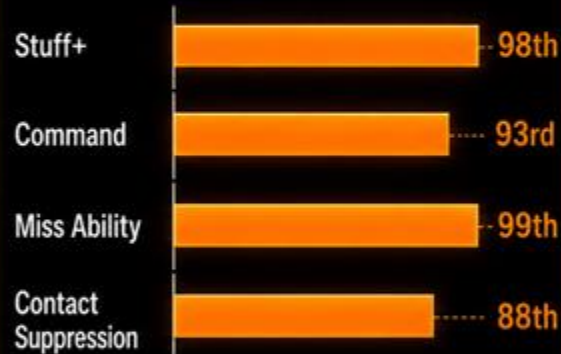
Case Study: What Eliteness Looks Like

Tarik Skubal

ERA: 2.21 | FIP: 2.45

Cy Young Winner

CDI SKILL PROFILE (PERCENTILE)



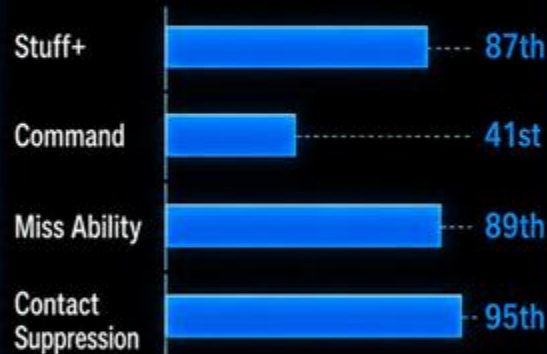
OVERALL CDI: **99.6th** PERCENTILE

Paul Skenes

ERA: 1.97 | FIP: 2.36

Cy Young Winner

CDI SKILL PROFILE (PERCENTILE)



OVERALL CDI: **89.9th** PERCENTILE

Tyler Rogers

ERA: 1.98 | FIP: 2.88

CDI SKILL PROFILE (PERCENTILE)



OVERALL CDI: **100th** PERCENTILE



CASE STUDY: GOOD CDI, POOR ERA/FIP

High underlying skill despite poor traditional run-prevention outcomes



PERCENTILE MISMATCH
CASE STUDY



SELECTION LOGIC Top CDI percentile + poor ERA/FIP percentile

Higher ERA/FIP percentiles correspond to lower ERA/FIP values.



30+ IP sample

SEAN MANAEA

93rd percentile CDI



TRADITIONAL RESULTS

ERA 5.64

12th pct

FIP 4.39

35th pct

PERCENTILE GAP

CDI% - Avg[ERA%, FIP%]

93.4 - Avg(12.3, 34.9)

+69.9

SKILL PERCENTILE RANKS

Stuff 66.9%

Command 68.0%

Miss 31.1%

Contact 60.1%

Why CDI liked him:

Command-driven profile with above-average stuff and contact.

CASEY LEGUMINA

90th percentile CDI



TRADITIONAL RESULTS

ERA 5.62

12th pct

FIP 4.38

36th pct

PERCENTILE GAP

CDI% - Avg[ERA%, FIP%]

90.1 - Avg(12.5, 35.5)

+66.1

SKILL PERCENTILE RANKS

Stuff 87.6%

Command 87.0%

Miss 13.5%

Contact 57.3%

Why CDI liked him:

Stuff-command profile despite weaker miss percentile.

TANNER SCOTT

86th percentile CDI



TRADITIONAL RESULTS

ERA 4.74

26th pct

FIP 4.70

26th pct

PERCENTILE GAP

CDI% - Avg[ERA%, FIP%]

86.0 - Avg(25.8, 25.6)

+60.4

SKILL PERCENTILE RANKS

Stuff 97.5%

Command 70.8%

Miss 98.7%

Contact 40.5%

Why CDI liked him:

Elite stuff and miss ability carried the profile.



Takeaway: CDI may identify underlying pitcher quality that ERA and FIP have not fully captured.

Application



Player Evaluation

(Identify true talent)

- Find undervalued pitchers
- Separate skill vs results
- Better scouting decisions



Player Development

(Pinpoint what to improve)

- what skill domain is lacking?
- Build targeted training plans
- track skill growth over time



Pitch Design

(Optimize pitch characteristics)

- improve miss ability
- shape pitches for weak contact
- align stuff with outcomes



Decision Making

(Support roster + strategy decisions)

- call-ups / demotions
- trade targets
- bullpen roles

Future Work

Next steps to strengthen and expand the CDI framework



1. EXPAND MODEL VALIDATION

Validate CDI across multiple MLB seasons and test year-to-year stability and predictive reliability.



2. IMPROVE SKILL WEIGHTING

Optimize skill weights using cross-validation and test variation by role or pitch mix.



3. INCREASE MODEL GRANULARITY

Add more context with location, splits, count, workload, and separate skills from environment.



4. BUILD A PRACTICAL ANALYTICS TOOL

Create an interactive tool to compare pitchers and deliver actionable insights for development.



GOAL: Refine the CDI framework and build tools that turn insights into better decisions and player development.

